Stage 1 Summary — Diagnostic Axis Model PRD

JEE Platform · agent-factory pipeline · iteration loop complete

1. What this stage was about

You asked me to research JEE Advanced past papers and design a question-component model so we can **pinpoint which kind of error** caused a student's wrong answer — not just whether they got it wrong, but *why* (reading mistake? counting mistake? case-handling? strategic bait?). Your verbatim request:

> “I want you to check 2026, 2025, 2022, 2020 jee advanced papers to build question layers. My main idea is to break a qs into several components, so that I can pin point reasons where a student is faltering and help him improve. It cud be topic based, comprehension based, reading mistake which are mostly because of a fixed variety of words, or counting mistake, or case driven mistake etc. so with each qs of PY you can associate time, difficulty, trickiness, etc and we can verify together to build a final model of question to axis for question”

2. How the agent-factory pipeline ran it

The agent-factory is a 5-stage pipeline. Every stage has a **generator** agent (who produces work) and a **discriminator** agent (who adversarially scores it). We loop until the discriminator scores the work 7/10 or higher, or until we hit 3 iterations and have to escalate to you.

Stage 1 (this stage) = Specification Loop. A Product Manager agent writes a Product Requirements Document (PRD). A Spec Critic agent adversarially reviews it. Score must hit 7/10 to advance to Stage 2 (Architecture).

The other four stages — not yet started — are: Architecture (Stage 2), Implementation (Stage 3), Testing (Stage 4), Integration / Ship (Stage 5).

3. Your clarifications (before agents started)

Before spawning any agents I asked you three focused questions so the PM had real constraints to work against:

Question	Your answer
Scope: which papers should the PM analyze?	2022 PCM only (Paper 1 + Paper 2). Future iterations can add 2023, 2024.
Axis relation: extend or refactor the existing 7-axis system?	Let the PM propose BOTH options, then I (and you) decide.
Years sanity check	“Recent + a few older” — add 2023/2024 if available.

4. Iteration history — the GAN loop in action

Three iterations. Each round: PM writes a PRD draft, Spec Critic scores it. If score < 7 we loop back. The score is supposed to go up.

Iteration	PM score	What changed	What went wrong
v1	7 / 10	First draft. Proposed 5 new axes orthogonal to existing 7.	6 blockers: count inconsistency (16 vs 18 questions), math didn't recon
v2	7 / 10	Fixed 4 of 6 v1 blockers. Tried to fix the DB invariant with Postgres GENERATED columns	STALLED. Same score as v1. Two blockers still partial
v3	8 / 10	Applied a META-FIX: stopped speccing implementation mechanisms. Delegated the DB invariant mechanism to St	ADVANCE. Score moved 7 to 8. No new blockers introduced. Three

Score trajectory: 7 → 7 → 8. v3 cleared the gate. Stage 1 complete.

5. What the final PRD actually proposes

After three iterations, the PRD landed on these decisions:

5.1 Add 5 new axes orthogonally

Your current 7-axis identity (TOPIC, SUBTOPIC, IDEA, SUB-IDEA, ANSWER-TYPE, SURFACE, TRAP) stays untouched. We **add** 5 new axes on top, total 12. The new axes describe *failure modes* — what kind of error a student typically makes on this problem — not the problem's character.

5.2 Tag per wrong-path, not per-problem

The 5 new axes are tagged **on each entry of wrong_paths** in the YAML file, not on the problem row directly. Why: a single problem usually has 2–3 wrong paths a student might take, and each path activates DIFFERENT failure modes. Tagging at the wrong-path level gives us per-mistake-type signal. Summary columns on the problem row aggregate this for fast query.

5.3 Two North Star metrics — one measurable now, one for pilot

- **Phase A (right now): Single-Path Match Rate (SPMR)** — for any student attempt, can we cleanly match their wrong answer to exactly ONE of the catalogued wrong paths? Target ≥ 90%. Computable against your existing 2 problems.
- **Phase B (after 20–50-student pilot): Diagnostic Success Rate (DSR)** — after a student sees the diagnosis we generated, does their next attempt on a similar problem succeed? Target ≥ 55%.

5.4 Honest reviewer-time budget

Phase	Time per question	Total cost
Calibration (first 30 questions, warm-up)	8–12 min	~5 reviewer-hours
Inter-rater set (2 reviewers tag same Qs)	—	~10 reviewer-hours
Steady state (after warm-up done)	≤ 4 min	ongoing
TOTAL up front		**~15 reviewer-hours**

5.5 DB invariant delegated to Stage 2 Architect

The summary columns on the problem row (e.g. failure_modes_seen[]) MUST stay consistent with the wrong_paths source-of-truth. The PRD does NOT specify HOW — it gives the Architect 5 acceptance criteria and lets them pick the mechanism (generated column, trigger, or materialized view). This was the meta-fix that broke v3 out of the v1/v2 stall: PM was trying to do architect work.

6. The 5 diagnostic axes — full value catalogue

19 axis values across 5 axes. Every wrong-path entry must pick exactly one value per axis. The all-NONE pattern is reserved for “didn't see the IDEA” (pure conceptual failure on the existing IDEA axis).

Axis	Value	What it tags
ERR-READING	NONE	No reading mistake on this path.
	QUANTIFIER	Misread “exactly / at least / contains / from each / non-zero / all”.
	NUMRANGE	Misread a numeric bound or interval endpoint.
	VARDEF	Misread a variable's defining property or domain.
ERR-CASE	NONE	No case-handling mistake.
	EDGE	Boundary or degenerate case dropped or double-counted (leading-zero, x=0, two params equ
	MODINDEX	Wrong period / parity / cycle length on iterated structure.
	PARTITION	Missing or duplicate sub-case in a case enumeration.
ERR-COMP	NONE	No computation slip.
	ARITH	Arithmetic slip (number added / multiplied wrong).
	SIGN	Sign or orientation flipped.
	ALGEBRA	Algebraic manipulation slip (wrong identity, factoring error).
ERR-STRATEGY	NONE	No strategy error.
	TRAP	Took the bait — invoked trap machinery when simpler method exists.
	PARTIAL	Multi-correct partial-marking miscalibration.
ERR-PARSING	NONE	Statement parsed correctly.
	LISTMATCH	Wrong List-I / List-II mapping (each computation right in isolation).
	GEOM3D	Failed to form / hold the geometric or 3D figure.
	PASSAGE	Misread the comprehension passage / multi-paragraph setup.

7. Two questions only you can answer (Stage 2 is paused on these)

The Spec Critic flagged these as decisions that must come from you, not from any agent. Stage 2 (Architecture) is parked until you've thought about them.

Q1: Are any failure modes missing from the 5 axes?

The PM analyzed 18 questions from JEE Advanced 2022 PCM and derived the 5 axes from what actually showed up. Your years of teaching may have seen failure modes those 18 questions didn't surface. Examples (use as prompts, not a checklist):

- Time-pressure / panic errors
- Formula recall failures (vs misapplication)
- Notation confusion (e.g. set vs sequence brackets)

- Conditional vs marginal probability conflation
- Unit / dimension errors (Physics)
- Mole / equivalent conversion errors (Chemistry)
- Visualization gaps that aren't 3D (graph sketching, vector mental rotation)
- NCERT-edge vs JEE-Advanced-edge conflation

Adding axes later means re-tagging the bank from scratch. Better to catch them now.

Q2: Who tags the 2022 calibration set?

15 reviewer-hours up front: ~5 for the warm-up calibration phase, ~10 for the two-person inter-rater set. Three sub-decisions:

- **Who are the 2 reviewers?** You + someone, or two external SMEs?
- **Who absorbs the cost?** Whose ~7.5 hours each does this come out of?
- **If only 1 available:** the student-facing diagnosis features stay feature-flag-gated (off) until a second reviewer arrives. Is that acceptable?

Two lower-priority decisions (can defer to Stage 2)

- **Q3:** Should reviewers write a free-text “evidence note” per tag? Cost: time. Benefit: explainability later. Default: no.
- **Q4:** When a student's wrong attempt doesn't match any catalogued path, CSV for periodic review or live queue UI? Default: CSV for v1, queue later.

8. What comes next

Stage 2 — Architecture Loop — PARKED

Once you answer Q1 and Q2, I spawn the Architect agent. Architect's job in Stage 2:

- Pick the DB-invariant mechanism (the choice delegated from the PRD).
- Add answer.precision field to the YAML schema (needed by the equality contract).
- Design the migration that extends the Prisma schema with the 5 new axes + summary columns.
- Integrate with existing taxonomy file structure.

Then Design Critic adversarially reviews. Loop to 7/10 or 3 iterations. Output: scorecards/02-architecture-final.md.

After Stage 2: Stages 3, 4, 5

Stage 3 = Engineer builds the migration + extends the importer. Stage 4 = Tester + UX Auditor verify it works. Stage 5 = Integrator runs end-to-end ship/no-ship gate. Each stage has its own generator/discriminator loop and its own scorecards.

9. Where everything lives on disk

File	What it is
`agent-factory/scorecards/01-prd-final.md`	The 700-line locked PRD (v3)
`agent-factory/scorecards/01-prd-draft-v1.md`	First PRD draft (7/10)
`agent-factory/scorecards/01-prd-draft-v2.md`	Second PRD draft (7/10, stalled)
`agent-factory/scorecards/01-prd-draft-v3.md`	Third PRD draft (8/10, advanced)
`agent-factory/scorecards/01-prd-review-v1.md`	Spec Critic v1 review (6 blockers)
`agent-factory/scorecards/01-prd-review-v2.md`	Spec Critic v2 review (4 new issues)
`agent-factory/scorecards/01-prd-review-v3.md`	Spec Critic v3 review (verdict: advance)
`agent-factory/scorecards/stage-1-summary.pdf`	This PDF
`_JEE PY Papers/2022_P1.pdf`	Input: JEE Advanced 2022 Paper 1
`_JEE PY Papers/2022_P2.pdf`	Input: JEE Advanced 2022 Paper 2
`docs/PROJECT CONTEXT.md`	Project binding spec (the 229-line constitution)
`agent-factory/CLAUDE.md`	Agent-factory constitution (the orchestrator's rules)